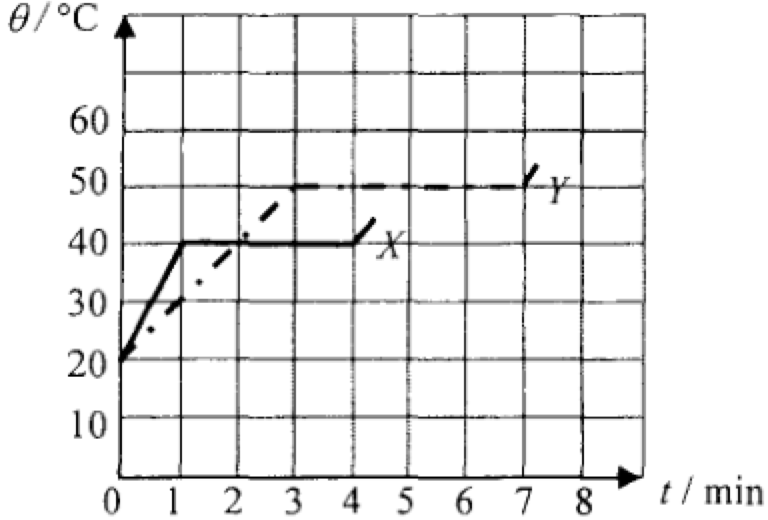
1. Solid substances X and Y of equal mass are heated by heaters of power $2P$ and P respectively. The graph shows how the temperature θ of each substance varies with the heating time t.



What is the ratio of the specific latent heat of fusion of X to that of Y ?

A. 3 : 2
B. 3 : 4
C. 4 : 3
D. 2 : 3

2. Metal blocks X and Y are identical in size and shape. X is of a higher temperature than Y. Which of the following statements must be correct ?

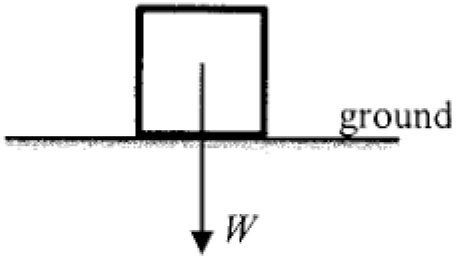
(1) Energy will flow from X to Y if they are in thermal contact.
(2) X is a better conductor of heat compared to Y.
(3) The total internal energy of X is higher than that of Y.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

*3. For an ideal gas, kinetic theory deduces that $pV = \frac{1}{3}Nmc^2$. Which physical quantity below can be represented by $\frac{3p}{\overline{c^2}}$?

A. the total mass of the gas
B. the volume of one mole of the gas
C. the density of the gas
D. the number of gas molecules per unit volume

4. A block of weight W is at rest on a horizontal ground as shown.



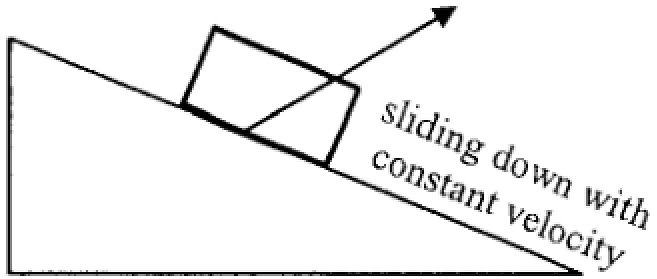
The force acting on the block by the ground is R. Which of the following statements is/are correct ?

(1) R and W are opposite in direction.
(2) R and W are equal in magnitude.
(3) R and W is an action-and-reaction pair.

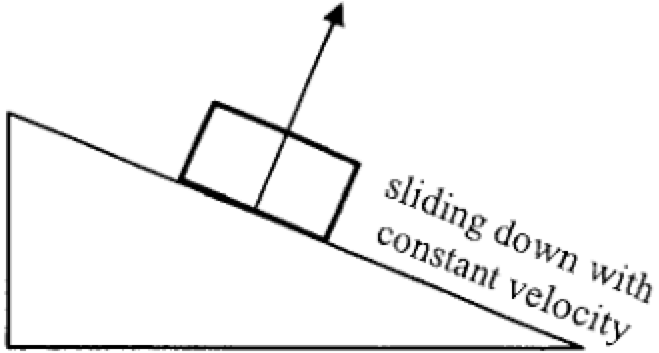
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

5. A block is sliding down a rough incline with constant velocity as shown. Which arrow indicates the direction of **the resultant force acting on the block by the incline** ? Neglect air resistance.

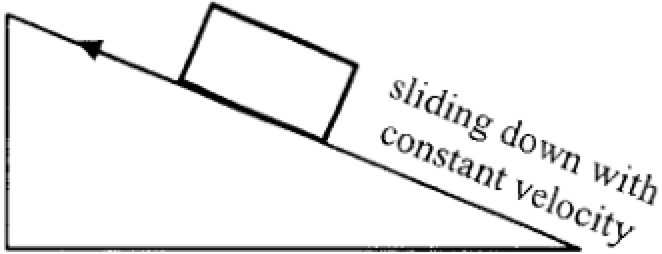
A.



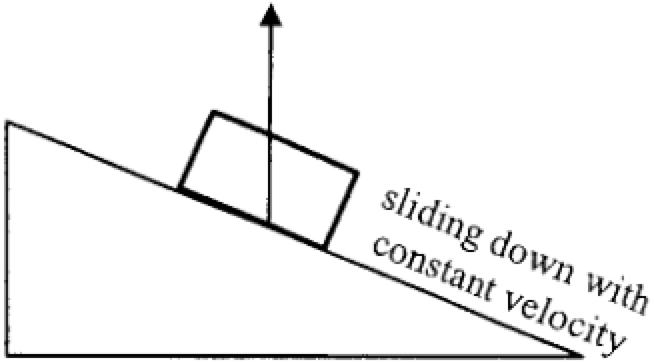
B.



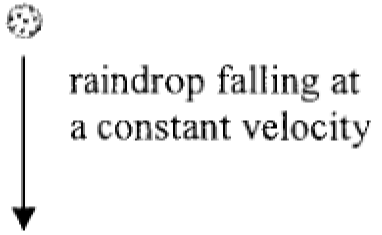
C.



D.

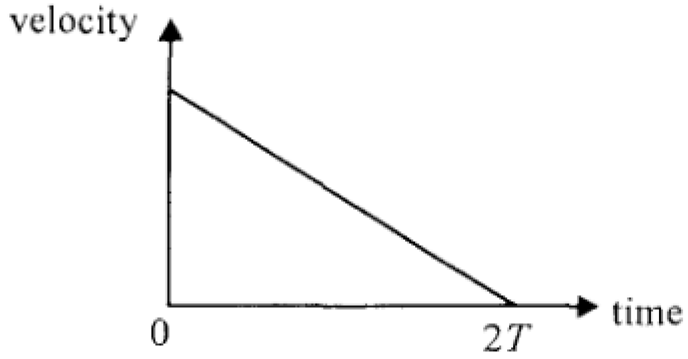


6. Which statement below about a raindrop falling at a constant terminal velocity is correct ?



A. No work is done on the raindrop by the gravitational force.
B. As the raindrop falls, all its gravitational potential energy loss is converted into kinetic energy gain.
C. The only force acting on the raindrop is its weight.
D. No net force is acting on the raindrop.

7. At time $t = 0$, a small sphere is projected up along a smooth incline with a certain initial velocity. It travels a distance L and becomes momentarily at rest after a time $2T$. The corresponding velocity-time graph is shown below.



What is the distance travelled by the sphere from $t = 0$ to $t = T$?

A. $\frac{1}{4}L$

B. $\frac{1}{2}L$

C. $\frac{3}{4}L$

D. $\frac{4}{5}L$

8. A stationary uranium nucleus ${}^{238}_{92}\mathrm{U}$ decays to give a thorium nucleus ${}^{234}_{90}\mathrm{Th}$ and an α particle ${}^{4}_{2}\mathrm{He}$.

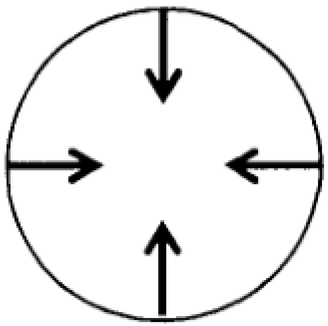
$$ {}^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} $$

Which of the following correctly describes the situation about the ${}^{234}_{90}\mathrm{Th}$ nucleus and the α particle just after the decay ?

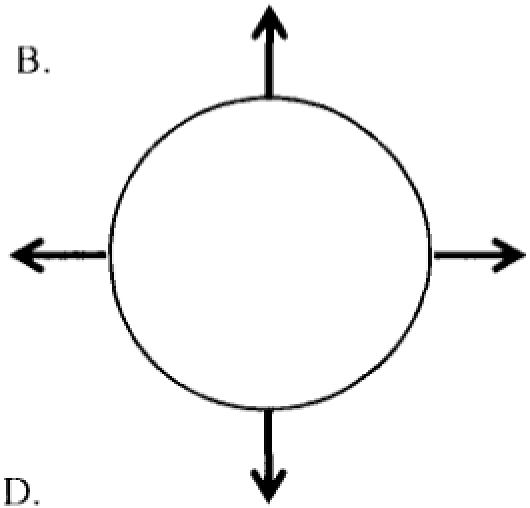
	magnitude of momentum p	kinetic energy KE
A.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} < \mathrm{KE}(\alpha)$
B.	$p(\mathrm{Th}) > p(\alpha)$	$\mathrm{KE(Th)} > \mathrm{KE}(\alpha)$
C.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} > \mathrm{KE}(\alpha)$
D.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} = \mathrm{KE}(\alpha)$

*9. A particle is moving clockwise (top view) in a horizontal circle with uniform speed. Which top view diagram below correctly shows the net force acting on the particle at various positions ?

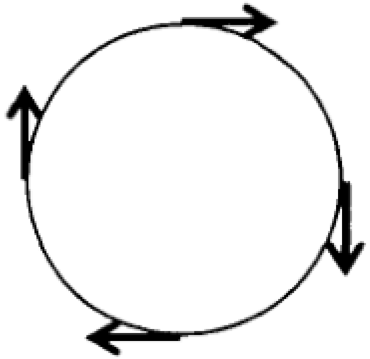
A.



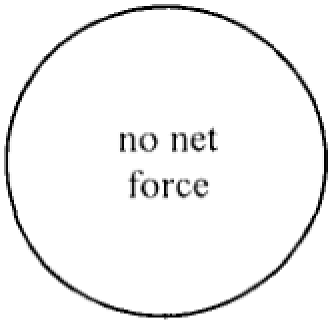
B.



C.



D.

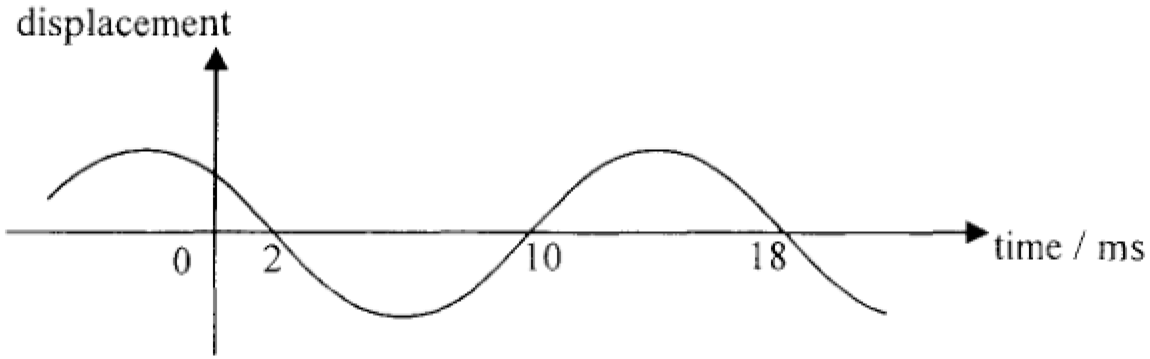


10. Which of the following can be transferred by mechanical waves from one place to another along the direction of propagation ?

(1) mass
(2) momentum
(3) energy

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

11.



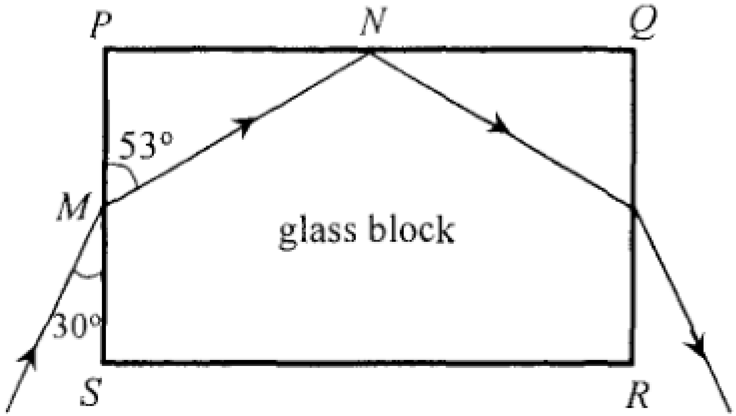
The displacement-time graph of a particle on a travelling wave is as shown. Find the frequency of the wave.

A. 55.6 Hz
B. 62.5 Hz
C. 111 Hz
D. 125 Hz

12. Earthquakes propagate in the form of waves. The quake centre produces both longitudinal waves (P-wave) and transverse waves (S-wave) which travel with speeds 9.6 km s^{-1} and 6.4 km s^{-1} respectively on the Earth's crust. In an earthquake, a monitoring station detects the arrival of the P-wave pulse at 7:02 a.m. while the S-wave pulse arrives 2 minutes later at 7:04 a.m. Estimate the time that this earthquake occurs.

A. 6:53 a.m.
B. 6:56 a.m.
C. 6:58 a.m.
D. 6:59 a.m.

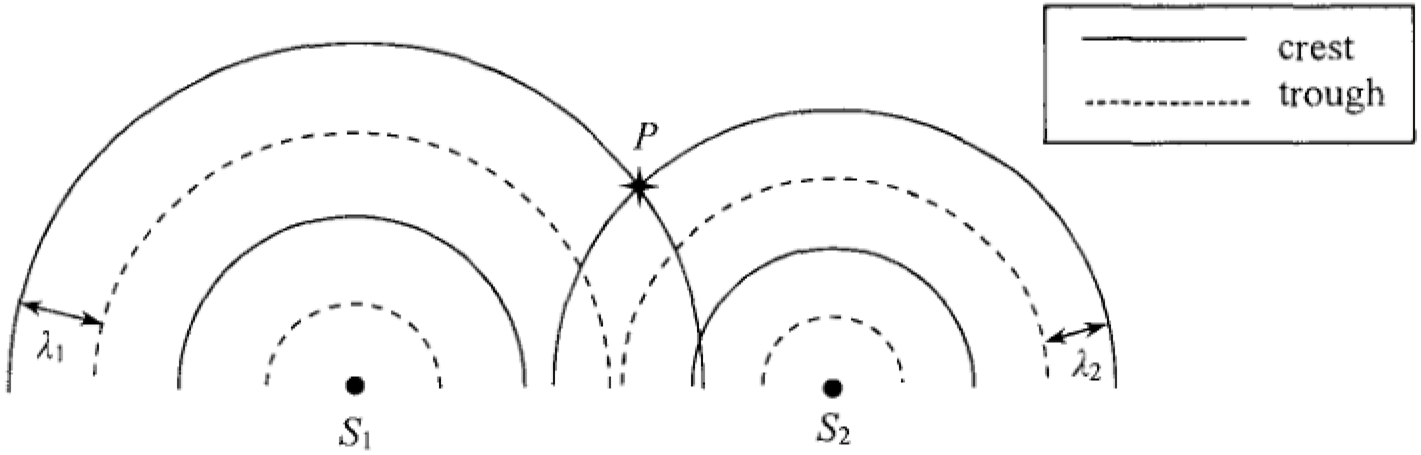
13.



The figure shows the cross-section of a rectangular glass block *PQRS*. A light ray is incident from air at *M* on face *PS* and the refracted ray strikes face *PQ* at *N*. Find the critical angle for the glass-air interface.

A. 37°
B. 44°
C. 53°
D. 60°

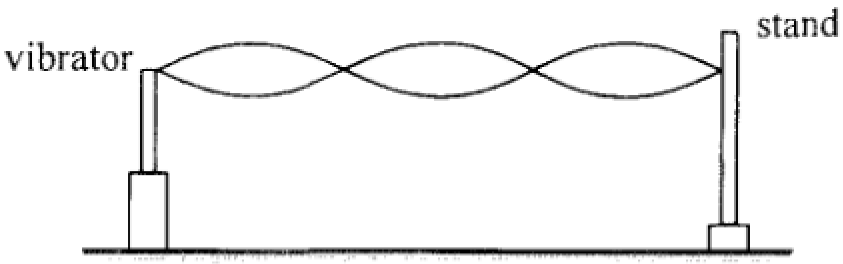
14. In a ripple tank, circular water waves of wavelengths λ_1 and λ_2 ($\lambda_1 > \lambda_2$) are produced by two vibrators S_1 and S_2 respectively. The figure represents the two circular waves propagating on the water surface at a certain moment.



Which of the following statements is correct ?

A. The particle at P is always at crest position.
B. At P, the two waves always reinforce to give a larger amplitude.
C. The principle of superposition cannot be applied at P as $\lambda_1 \neq \lambda_2$.
D. The principle of superposition can be applied at P but the two waves do not always reinforce at that location.

15. In the set-up below, different stationary wave patterns are formed on an elastic string by adjusting the frequency f of the vibrator.

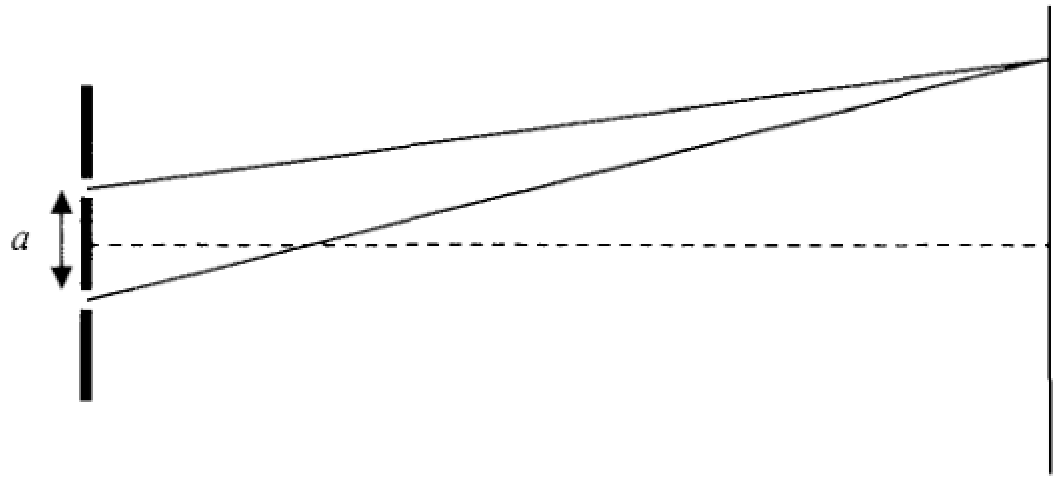


Which statements are correct when f increases ?

(1) The number of antinodes increases.
(2) The speed of the waves on the string increases.
(3) The frequency of the waves produced in air by the string increases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

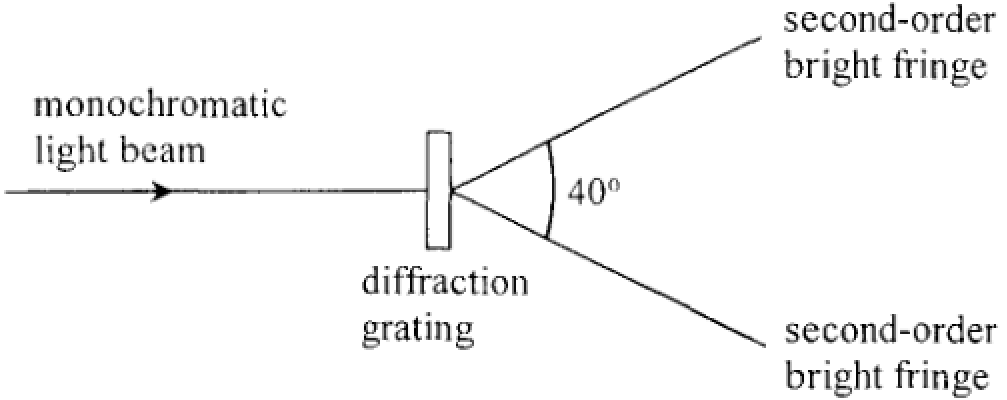
16. In Young's double slit experiment employing monochromatic light, how will the interference pattern change if the separation a of the double slit is reduced ?



(1) The pattern will become brighter.
(2) The number of fringes that can be observed will increase.
(3) The fringe separation of the pattern will become larger.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

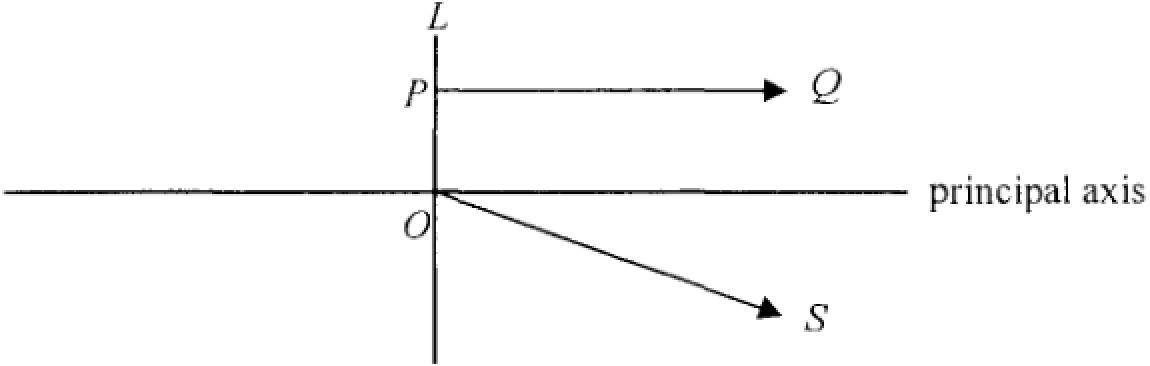
*17.



When a monochromatic light beam is incident normally on a diffraction grating with 300 lines per mm, a pattern of bright fringes is formed. The angle between the two second-order bright fringes is 40°. Find the frequency of the light.

A. 1.4×10^{14} Hz
B. 2.6×10^{14} Hz
C. 2.8×10^{14} Hz
D. 5.3×10^{14} Hz

18. In the figure below, PQ and OS are light rays refracted from a lens L. Both light rays come from a point object situated on the left of L.



Which of the following deductions is/are correct ?

(1) The image of the object must be virtual.
(2) The object must lie along the straight line containing OS.
(3) L must be a concave lens.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

19. Typical wavelengths of X-rays and microwaves are in the ratio $10^n : 1$. The value of n could be

A. −10.
B. −4.
C. +4.
D. +10.

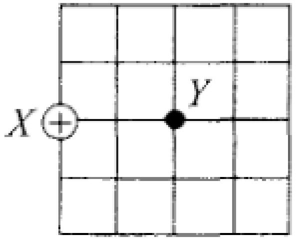
20. Submarines employ ultrasound instead of microwaves to detect obstacles in the sea. This is because

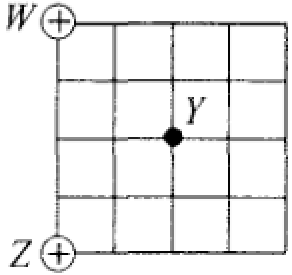
A. wavelengths of ultrasound are shorter than those of microwaves.
B. ultrasound travels faster than microwaves in the sea.
C. microwaves are easily absorbed by sea water.
D. microwaves diffract too much in the sea.

21. Three isolated identical metal spheres X, Y and Z carry charges $+2Q$, $-4Q$ and $+5Q$ respectively. Y is first moved to touch X and then Y is brought in contact with Z. When Y and Z are separated, find the charge on each sphere.

	X	Y	Z
A.	0	$+1.5Q$	$+1.5Q$
B.	$-Q$	$+0.5Q$	$+0.5Q$
C.	$+Q$	$+Q$	$+Q$
D.	$-Q$	$+2Q$	$+2Q$

*22. When a point charge $+Q$ is placed at X as shown, the strength of the electric field at Y is E_0.





If W and Z are each placed with a point charge of $+Q$, what will be the electric field strength at Y ?

Note: $\sin 45° = \cos 45° = \frac{\sqrt{2}}{2}$

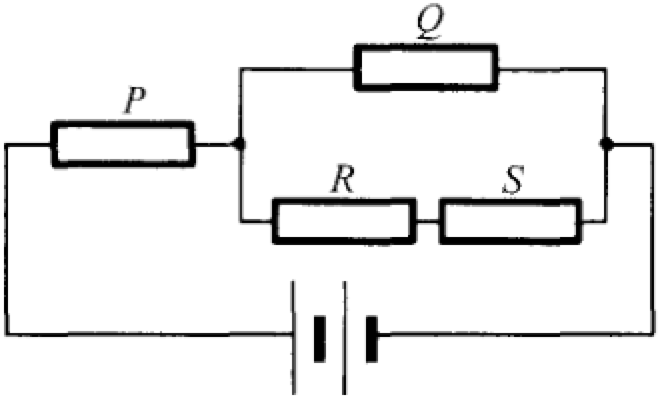
A. $\frac{\sqrt{2}}{2} E_0$

B. E_0

C. $\sqrt{2}\, E_0$

D. $2\, E_0$

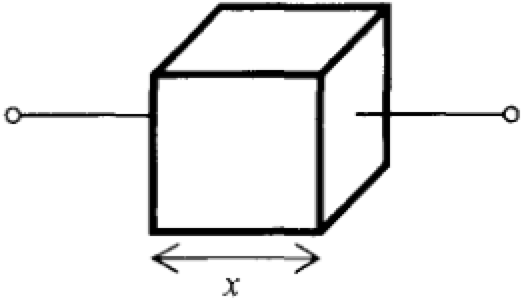
23. Four identical resistors P, Q, R and S are connected to a battery of negligible internal resistance as shown.



If the power dissipated by R is 1 W, find the total power output of the battery.

A. 11 W
B. 15 W
C. 19 W
D. 21 W

24. The figure shows a metallic cube of side length x. How is its resistance R between any two opposite faces related to x ?



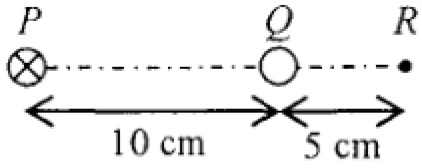
A. $R \propto \frac{1}{x}$

B. $R \propto x$

C. $R \propto x^2$

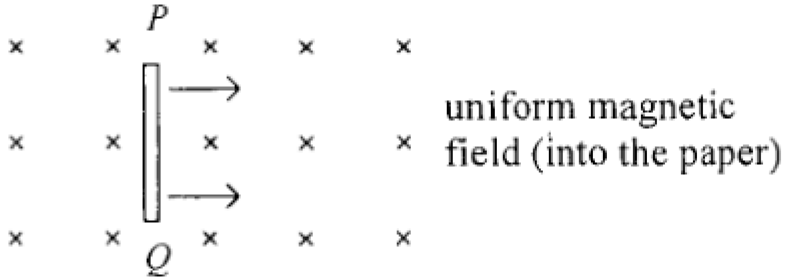
D. $R \propto \frac{1}{x^2}$

25. In the figure below, PQR is a straight line with $PQ = 10$ cm and $QR = 5$ cm. A long straight wire carrying a current of 0.3 A (directed into the paper) is placed at P. When another long straight wire carrying a current I is placed at Q, the resultant magnetic field at R becomes zero. Determine the direction and magnitude of I.



	direction of I	magnitude of I
A.	into the paper	0.1 A
B.	into the paper	0.9 A
C.	out of the paper	0.1 A
D.	out of the paper	0.9 A

26. When a copper rod PQ moves with a constant velocity across a uniform magnetic field as shown, an e.m.f. is induced across the rod.

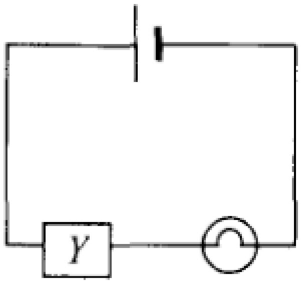


Which of the following statements is/are correct ?

(1) The magnitude of the induced e.m.f. depends on the length of the rod.
(2) Rod PQ acts like a cell providing an e.m.f. with P being its positive terminal.
(3) There is a force acting on the rod to oppose its motion.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

27. A light bulb is connected in series with a device Y and a cell as shown. Assume that the internal resistance of the cell is negligible and its e.m.f. remains unchanged.

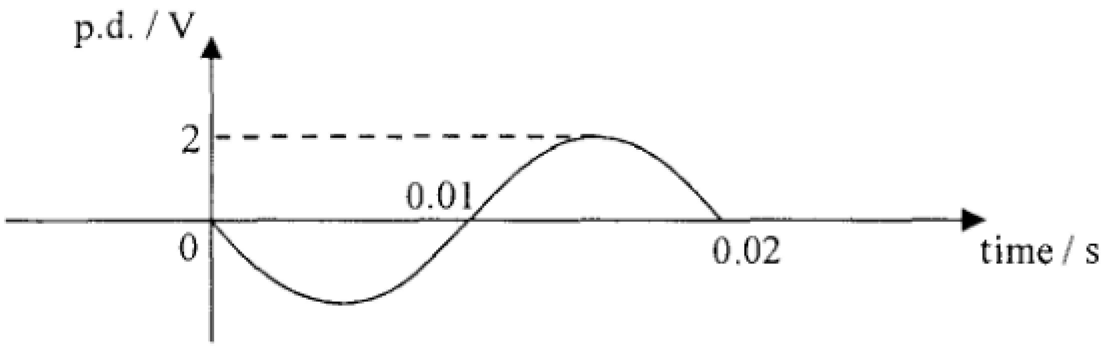


It is found that the brightness of the light bulb decreases with time. Which deductions must be correct ?

(1) The current in Y decreases with time.
(2) The voltage across Y decreases with time.
(3) The power supplied by the cell decreases with time.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

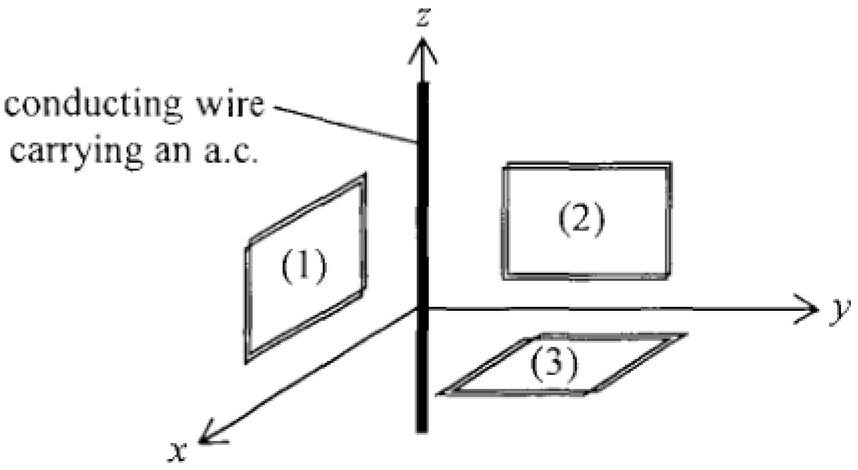
*28. The graph shows the waveform of a sinusoidal alternating p.d. applied across a 10 Ω resistor.



Find the root-mean-square current in this 10 Ω resistor and the average power dissipated by it.

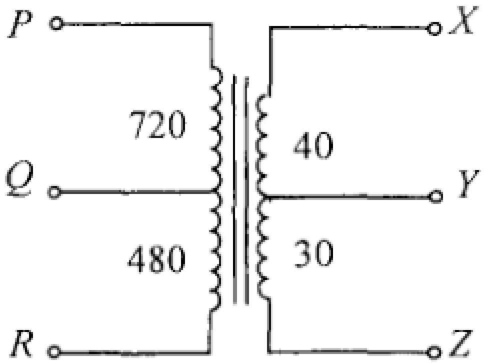
	root-mean-square current / A	average power / W
A.	0.14	0.2
B.	0.14	0.4
C.	0.2	0.2
D.	0.2	0.4

29. The figure shows three mutually perpendicular coils (1), (2) and (3) placed near a conducting wire carrying an a.c. along the z-axis direction. In which coil(s) would e.m.f. be induced ?



A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

*30.



The above figure represents a multi-tapped transformer. The number of turns between the various 'tapping points' are indicated as shown. Which connections should be used for stepping down a voltage from 240 V to 6 V ?

	primary coil	secondary coil
A.	PQ	XY
B.	PQ	YZ
C.	PR	XY
D.	PR	YZ

31. A radioactive nuclide plutonium-239 ($^{239}_{94}Pu$) becomes a stable lead-207 isotope ($^{207}_{82}Pb$) after a series of α and β decays. Find the number of β decays in the process.

A. 3
B. 4
C. 5
D. 6

32. The activity of a radioactive sample is measured to be 18 MBq. What is its activity 3 half-lives before ?

A. 6 MBq
B. 54 MBq
C. 72 MBq
D. 144 MBq

33. Which of the following may contain sources of ionizing radiatio

(1) sea water
(2) a rock sample
(3) human body

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)